



SMM921: Trading and Microstructure

Quantitative Investing: Black-Litterman, Costs, and Dynamic Allocation

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Portfolio construction and transaction costs

In this lecture topic we will cover:

- The basics of portfolio construction
 - Pitfalls
 - Solutions: Bayesian stats, applying weight constraints [including video lecture]
- Incorporating trading costs into portfolio construction
- Introducing dynamic asset allocation

Reading

Look at:

- Ang, Chs. 3 and 4
- Grinold and Kahn, Chs. 14, 15 and 16
- Grinold and Kahn, “*Advances in Active Portfolio Management*”, Ch. 6
- Kinlaw, Kritzman and Turkington, Chs. 5, 8 and 15

Startpoint: linear mean-variance optimization

We know this well by now. Choose weights (w) to achieve the following:

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \quad s.t. \quad \mathbf{1}'\mathbf{w} = 1$$

Thus:

- Most commercial optimizers are based on this linear mean-variance paradigm
- λ is a constant and equal to investor risk aversion
- $\boldsymbol{\alpha}$ is our set of return forecasts and $\boldsymbol{\Sigma}$ is the return covariance matrix

Solution to the linear mean-variance problem

For constant λ , we can derive an **explicit solution** to the linear mean-variance problem. A small amount of calculus delivers:

$$\mathbf{w} = \frac{1}{\lambda} \Sigma^{-1} \alpha$$

Intuitively:

- Assets with larger return variances tend to have smaller (absolute) weights
- Assets with larger expected returns tend to get bigger weight

Solution to the linear mean-variance problem

If we also allow the manager to invest in a risk-free asset with return r_f then we can reformulate the linear problem as follows:

$$\max_{\mathbf{w}} \mathbf{w}'\alpha + (1 - \mathbf{w}'\mathbf{1})r_f - \frac{\lambda}{2}\mathbf{w}'\Sigma\mathbf{w} \quad (1)$$

where the risky-asset weights need not sum to one, since the risk-free asset ensures total weights do. The solution is:

$$\mathbf{w} = \frac{1}{\lambda}\Sigma^{-1}(\alpha - \mathbf{1}r_f)$$

Note: it is now obvious that when the investor becomes more risk-averse (i.e, λ increases) portfolio weights for all risky assets get shrunk towards zero

Estimation error and ‘error maximization’

Key problem:

- α and Σ are both forecasts and subject to **estimation error**
- As we discussed in the last lecture, this estimation error distorts optimized portfolios
- We called this problem **error maximization**

Recap of intuition:

- Assume a PM has estimates of α and Σ
- The optimizer tends to take extreme long/short positions from small mean-return errors
- Extreme estimated correlations can also produce “corner solutions”
- Weights then become overly dispersed and the PM may trade noise rather than signal

Note: For more content, see Jupyter Notebook exercise ‘MV_with t_costs_L6’

Estimation error and ‘error maximization’

Key references:

- Michaud (1989, FAJ) “The Markowitz Optimization Enigma: Is Optimized Optimal”
- Best, M. J., and Grauer, R. R. (1991, RFS) “On the Sensitivity of Mean-Variance-Efficient Portfolios to Changes in Asset Means: Some Analytical and Computational Results”
- Chopra, V. K., and Ziemba, W. T. (1993, JPM) “The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice”
- De Miguel, Garlappi and Uppal (2007, RFS) “Optimal versus naive diversification: How inefficient is the $1/N$ portfolio strategy?”

Remedies

- Arguably, the **easiest** solution is to **set constraints** to avoid corner solutions:
 - U.S. equities are ~65% of the total equity market today
 - Constrain the portfolio to have U.S. equity between say ± 15 percentage points from market (i.e., 50-80%)
 - Is a 15-point deviation a lot?
- Other **common alternatives** include:
 - **Shrinkage** (e.g., Black-Litterman)
 - **Robust optimization** (e.g., $\max\{\min\{\}\}$)
 - **Resampling techniques** (e.g., Resampled Efficient Frontier (REF) by Michaud, 1998)

The Black-Litterman model

Startpoint: mean-variance optimal portfolios can be reverse-engineered such that we can have an estimate of what, in aggregate, the “market” is expecting from the market portfolio

- **Mean-variance framework:** use return predictions (α) to build the optimal portfolio

$$\mathbf{w} = \frac{1}{\lambda} \Sigma^{-1} \alpha$$

- **Implied market returns:** infer the market returns that are consistent with a particular market portfolio (π) by reverse engineering

$$\pi = \lambda \Sigma \mathbf{w}_{mkt}$$

Note: For more content, see Jupyter Notebook exercise ‘Implied_BL_returns_L6’

The Black-Litterman model

Black-Litterman is a bayesian approach that combines:

- prior market-implied returns
- investor views
- uncertainty around both

Starting with the prior on expected returns:

$$\mu \sim N(\pi, \tau\Sigma)$$

where τ is a scalar controlling uncertainty in the prior

The Black-Litterman model

Black–Litterman **posterior expected returns** become:

$$\mu^{BL} = \pi + \tau \Sigma \mathbf{P}' (\mathbf{P} \tau \Sigma \mathbf{P}' + \Omega)^{-1} (q - \mathbf{P} \pi)$$

where $\mathbf{P}$ is the pick matrix telling which assets each view refers to, q is vector of view returns, and Ω is the covariance matrix of view errors

Interpretation: smaller/larger Ω leads to more/less confidence in views

Intuition:

- start from market equilibrium π
- tilt toward the views q
- tilt is stronger when confidence is higher

Once posterior returns are obtained, you can compute optimal mean-variance weights

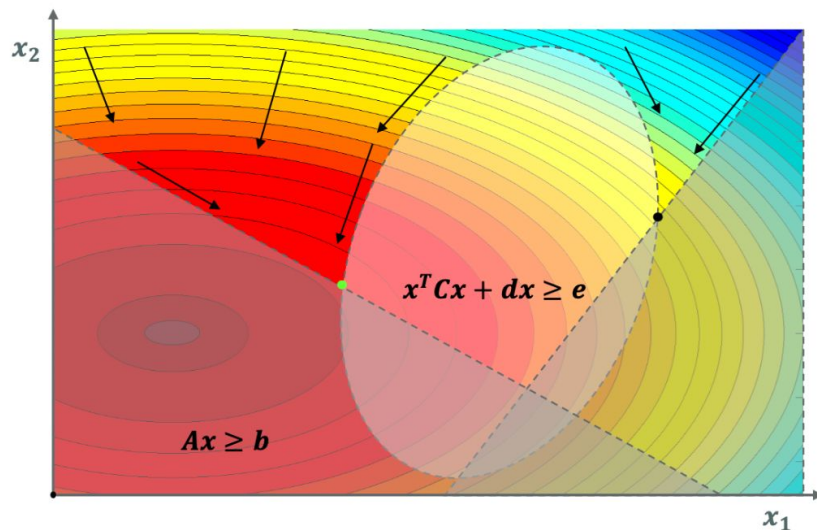
The importance of portfolio constraints

Reasonable constraints are essential for portfolio construction and reaching the **global optimum**, but some make the problem much harder

Even with concave utility functions such as quadratic or CRRA, **non-convex constraints** can produce **local optimum**

Examples:

- Mixed-integers constraints
 - (e.g., fix trading costs)
- Upper bound quadratic constraints
 - (e.g., portfolio volatility)
- Median vs. mean weight constraints



Portfolio management and t-costs

The choice of portfolio weights should also depend on **transaction costs**. Intuitively, a portfolio manager should prefer assets that are cheaper to trade, other things equal

Penalised MV optimisation: consider the augmented mean-variance problem below:

$$\max_{\mathbf{w}} \mathbf{w}'\alpha - \lambda \mathbf{w}'\Sigma\mathbf{w} - \tau' |\mathbf{w} - \mathbf{w}_0|$$

where:

- $\mathbf{w}$ are weights and $\mathbf{w}_0$ is your starting portfolio weight vector
- α is a vector of expected returns
- Σ is a return covariance matrix
- λ is the risk aversion
- τ is a vector of t-costs

Comments on the augmented MV problem

With our modified objective function:

- We need to measure t-costs before we form portfolio weights
- The penalty term will discourage trading in stocks for which the corresponding element of τ is large
- There is now no closed form solution (because of the absolute value term) but it is still convex

Fundamental question: having measured τ , how do we arrive at an optimal portfolio?

Optimization with t-costs

Basic problem:

- The key problem when t-costs are involved in the optimization is the fact that they depend on the **absolute value** of portfolio weight changes
- The optimization problem is no longer quadratic and non-smooth

We can come to an **approximate solution** to the problem, obtainable using quadratic optimization:

$$\tau' |\mathbf{w} - \mathbf{w}_0| = \mathbf{c}'(\mathbf{w} - \mathbf{w}_0) \text{ where } c_i \begin{cases} \tau_i & \text{if } w_i > w_{0,i} \\ -\tau_i & \text{if } w_i \leq w_{0,i} \end{cases}$$

Note:

- Now the penalty term appears to be (but is not really) linear in $\mathbf{w}$
- Our t-cost vector is transformed into $\mathbf{c}$, a function of τ and the final portfolio weight $\mathbf{w}$

Optimization with t-costs

An approximate solution:

1. First solve the zero t-cost problem:

$$\max_{\mathbf{w}} \mathbf{w}'\alpha - \lambda \mathbf{w}'\Sigma\mathbf{w}$$

Denote the solution to this problem $\mathbf{w}^*$

2. Define the t-cost vector, $\mathbf{c}$, as follows:

$$c_i = \begin{cases} \tau_i & \text{if } w_i^* > w_{0,i} \\ -\tau_i & \text{if } w_i^* \leq w_{0,i} \end{cases}$$

Optimization with t-costs

3. Solve with costs and with the constraint that the change in a stock's weight must be of the same sign as the change in weight from $\mathbf{w}_0$ to $\mathbf{w}^*$

$$\max_{\mathbf{w}} \mathbf{w}'(\alpha - \mathbf{c}) - \lambda \mathbf{w}'\Sigma\mathbf{w}$$

subject to the additional constraints:

$$\begin{cases} w_i > w_{0,i} & \text{for those } i \text{ where } w_i^* > w_{0,i} \\ w_i \leq w_{0,i} & \text{for those } i \text{ where } w_i^* \leq w_{0,i} \end{cases}$$

Comment: this two-step solution is only approximate but is sufficiently accurate to serve in most cases

Practical application: t-costs in optimization

Take our momentum trading example. Embed some t-costs into that problem:

$$\max_{\mathbf{w}} \mathbf{w}'\alpha - \lambda \mathbf{w}'\Sigma\mathbf{w} - \tau' |\mathbf{w} - \mathbf{w}_0|$$

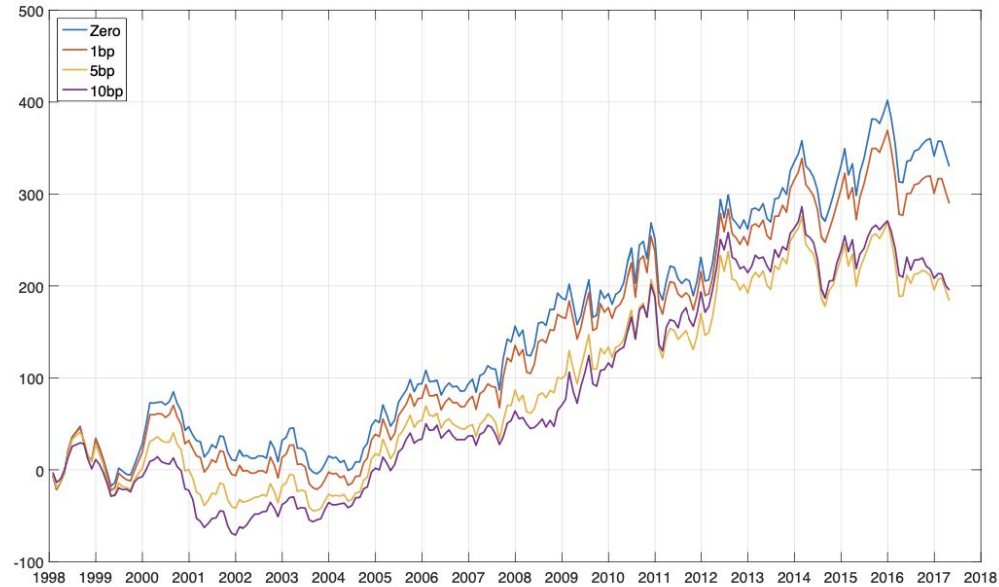
Keep all the parameters the same as last week except:

- Run the model four times; for trading costs of 0.00%, 0.01%, 0.05% and 0.10% (costs are assumed equal for all stocks)
- Also account for trading costs in computing the total return each month. Returns equal:

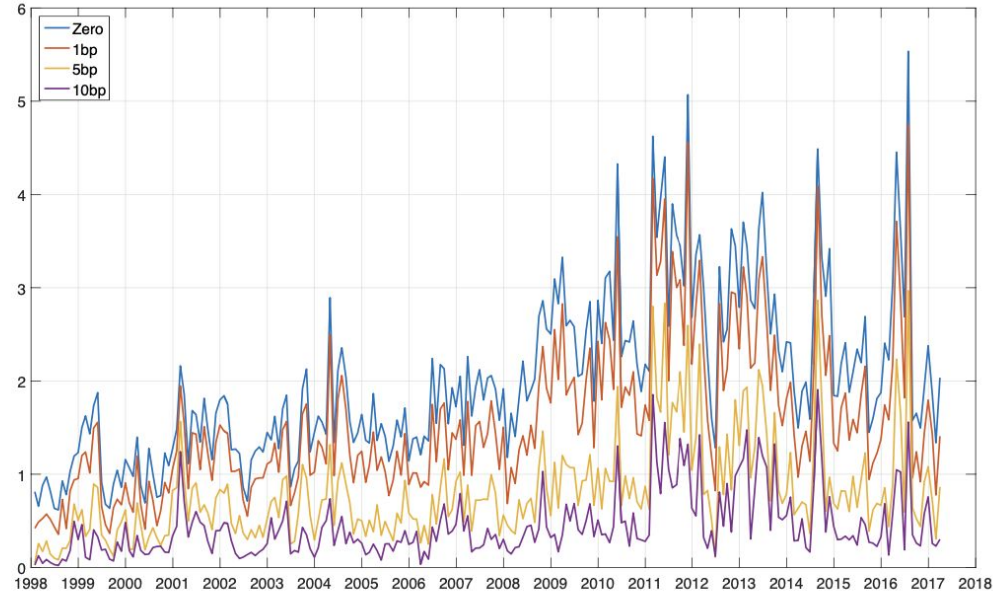
$$r_{p,t+1} = \mathbf{w}_t' \mathbf{r}_{t+1} - \tau' |\mathbf{w}_t - \mathbf{w}_{t-1}|$$

Note: For more content, see Jupyter Notebook exercise 'MV_with t_costs_L6'

Example: cumulative returns for various t-cost assumptions



Example: turnovers for various t-cost assumptions



Example: portfolio stats across t-cost assumptions

Observations:

- Trading costs obviously tend to reduce mean returns and reduce performance as measured by IR
 - Because you take more conservative positions
 - And costs directly erode P&L
- But the more conservative positions leads to reduced risk and greatly reduced turnover
- Sometimes, reduced **turnover** can improve performance!

Example: portfolio stats across t-cost assumptions

Limitations:

- What's the right t-cost assumption?
 - Clearly zero is unreasonable
 - Is 0.1% reasonable? What does 0.1% mean?
- Surely costs should vary across stocks
 - Small caps expensive to trade
 - Blue chips cheap to trade
- Costs will also vary over time
 - Larger in more volatile periods and lower volume periods
 - Why?
 - What does the answer to the 'why' question imply about how we should model and forecast t-costs?

Other extensions to the standard linear problem

The standard linear mean-variance problem is mathematically tractable, but it is limited from a practical perspective and not very robust. Some standard modifications:

- **Short-sales constraints:** many portfolio managers can only take long positions and so portfolio weights for such agents must be restricted to be positive::

$$\max_{\mathbf{w}} \mathbf{w}'\alpha - \lambda \mathbf{w}'\Sigma\mathbf{w} \text{ s.t. } \mathbf{w} > 0$$

- **Maximum positions:** impose upper bounds on weights, so as to ensure a fairly large degree of diversification. Helps deal with error maximisation:

$$\max_{\mathbf{w}} \mathbf{w}'\alpha - \lambda \mathbf{w}'\Sigma\mathbf{w} - \tau' |\mathbf{w} - \mathbf{w}_0| \text{ s.t. } \mathbf{w} > 0, \mathbf{w} < \bar{\mathbf{w}}$$

Where $\bar{\mathbf{w}}$ is a vector of upper bounds on portfolio weights

Other extensions to the standard linear problem

- **Turnover constraints:** many managers attempt to limit the amount of trading they do during any particular day/week/month to save money on trading costs, limit portfolio churn and to limit their visibility to other market participants:

$$|\mathbf{w} - \mathbf{w}_0| < K$$

- **Industry/market exposure management:** some managers impose limits on the total portfolio weight one might devote to a particular industry/market/segment

These complications introduce multiple equality and inequality constraints (non-convex!) into the problem. As a result, the problem no longer admits a closed-form analytical solution

Alternative frameworks

The portfolio construction approach outlined above rests on well-established and consistent theoretical foundations and is widely applied in practice. However, it is by no means the only framework in use. Alternatives include:

1. Risk budgeting:

- To maximise portfolio expected return subject to a constraint on overall risk
- Risk may be measured by portfolio return variance, VaR or some other measure
- Note this is not standard budgeting, as risks are not additive

2. Stratification:

- Divide stocks into distinct, non-overlapping groups, for example by industry
- Select stocks within each group in line with the investor's preferences
- The selection criterion would be based on a risk-adjusted expected return measure
- If desired, stocks can also be grouped along multiple dimensions rather than just one

Alternative frameworks

3. Factor exposure targeting:

- Decide on a set of factors that one believes adequately describe security risk
- Maximise expected return subject to the constraints that your portfolio factor exposures are within certain limits or at a certain level
- For example, if you like the CAPM, one could construct a portfolio by maximizing expected return subject to the constraint that the portfolio's β was zero

An introduction to dynamic asset allocation

Imagine you can build a portfolio composed of two assets:

- Risky asset with expected return 5%, volatility 15%, and **mean-reverting** returns
- Risk-free asset with return 5%

Long-term investors using **dynamic asset allocation** would allocate a small amount to the risky asset. This is called “**hedging demand**”

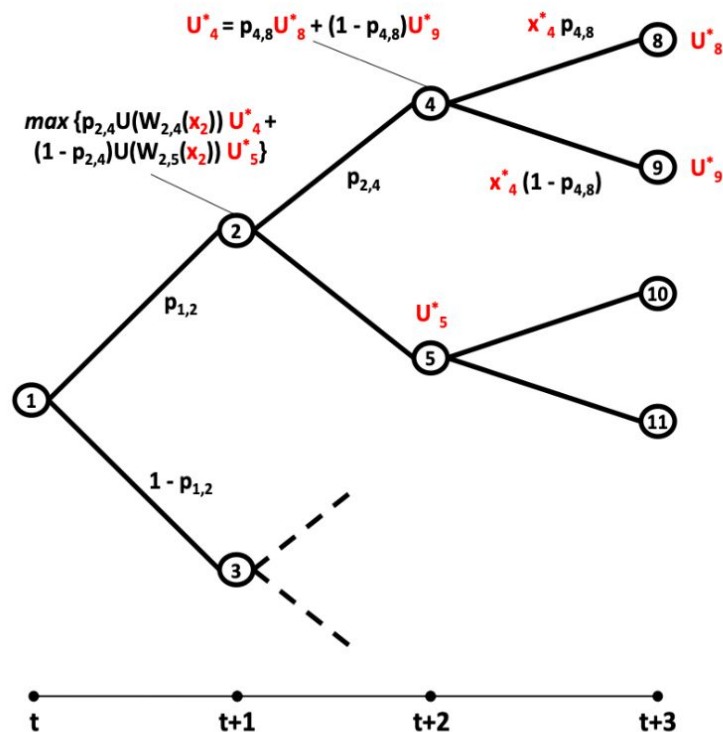
The fact the risky asset has ‘**mean reversion**’ means:

- If next period is a “good” state, high confidence the risky asset will perform well in the future
- In the “bad” state with a poor investment outlook, it provides higher wealth (hedge)

Backward induction and dynamic allocation

Using **dynamic programming** and **backward induction**:

- Starting at $t + 2$ the investor chooses the weights to maximize the expected utility at $t + 3$
- These conditional utilities are known as “indirect” utilities
- By following an iterative process, solve for the full set of optimal weights up to time t



Backward induction and dynamic allocation

Difficulties with tree/lattice approach:

- Multiple branches: problem quickly becomes enormous (i.e., curse of dimensionality)
- Defining p (or defining up/down states)

Can overcome both problems by combining Monte Carlo simulations and OLS regressions

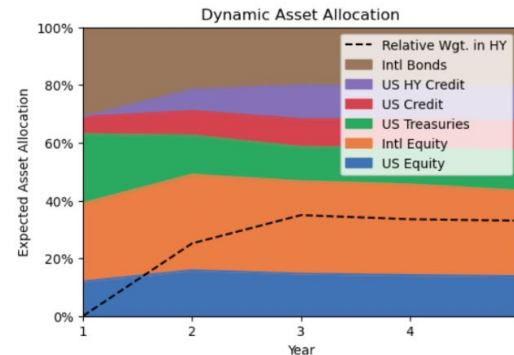
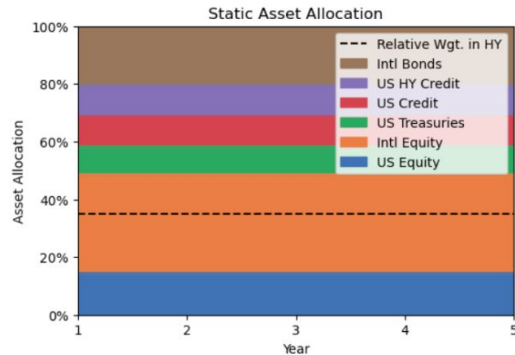
- This is known as **Least-Squares Monte Carlo**

Note: For more content, see Jupyter Notebook exercise 'DP_Markov_model_L6'

Example: Are 'junk bonds' junk for a portfolio?

Static model allocates significant portion to HY, whereas dynamic model does not.
Why?

- Tight spreads (expectation: loosening in the next few years)
- High inflation (HY firms cannot pass input costs onto consumers → profit margin compression → higher risk of financial distress)



Notes: Spread is defined as the difference in yield between the Bloomberg US Corporate High Yield Bond Index and the US 5-Year Treasury. The relative weight in high-yield is computed as US High-Yield Credit divided by total US Fixed Income. Constraints were total equity between 30% and 70%; equity home bias between 30% and 75%; bond home bias between 45% and 70%; US Treasuries less than 80% of total US fixed income; US investment-grade debt (US Treasuries plus US Credit) at least 50% of US fixed income.

Summary

We have discussed:

- Problems associated with the **standard linear problem** due to measurement error
- A Bayesian solution to this problem: **Black-Litterman**
- Introduced **transaction costs** to our problem to make it more realistic and saw how our solutions change
- We have also discussed some **alternative approaches** to the classical paradigm
- Finally we introduced the principle and key techniques used for expanding the model to a **multi-period setting** and act optimally through time

Video lecture: you will see how t-costs can be embedded in outcomes and how they affect optimal portfolios and their returns